

Test protocol

Connect the clients to the internet.

Power the clients one by one.

Open rpicam on each camera → adjust the focus.

 Disconnect Wi-Fi before unplugging anything.

Connect all Ethernet cables without powering the devices.

Power on the Master and connect it to the internet.

Power on the clients and wait about 1 minute for the Raspberry Pis to boot.

Set the button to the ON position.

Tag at the beginning of the video.

Note the exact tag time (including milliseconds).

Perform a test lasting at least 10 minutes (ideally 20 minutes).

Tag again before stopping the recording.

Note the exact tag time (including milliseconds).

→ The two tags make it possible to determine the delay between the recorded tag and the actual tag, to verify that the delay is minimal.

Set the button to the OFF position.

To start a new recording, wait until the LED has stopped blinking.

In the log.txt file:

Retrieve the tag indices.

Retrieve the frame corresponding to each index (using the program **Frames_Opti.py**).

Check that the frame matches the tag (perform the test by filming a stopwatch).

If the tag is delayed by more than 1 s → there is a problem.

Possible cause : poor-quality Ethernet cables.

Repeat the test with shorter videos (about 2 minutes).

Tag at both beginning and end of each video.

Record about 3 videos in a row (always paying attention to the LED status).

Check that there is no mix-up between videos.

Check that the start time of the video matches the first time in the info.txt file.

Once the system is functioning correctly, do not unplug either the clients or the master.

Vocabulary

Tag = press the tag button.

Rpicam = to open in a terminal using the command :
rpicam-hello --timeout 0

Internet Connection

Always connect through a mobile hotspot..

Do not use the Wi-Fi network KTHOPEN.

LED Usage

While the LED is blinking, never tag or press the button.

If the LED is steady, a recording is in progress.

Viewing the Raspberry screen on a laptop

Connect only the master to the internet.

Install RealVNC Viewer on the laptop.

Enter the master's IP address (to be retrieved on the phone in the connection sharing tab)